

# Yahya Zakaria

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## Education

**Bachelor of Computing and Data Science**, Alexandria University 2023 – 2027  
• **CGPA: 3.9839/4.0** | **Achievements:** Ranked 1st in department

## Experience

**Software Engineer Intern**, Deloitte Innovation Hub – Cairo, Egypt Jul 2026 – Aug 2026  
• Designed and implemented an internal **CV search system** using **Spring Boot**, **Python**, **RabbitMQ**, **PostgreSQL**, and **Vertex AI**, processing batches of **50 CVs**.  
• Built the **Dockerized** environment and **CI/CD pipelines**, and resolved critical issues across the end-to-end processing pipeline.  
• Collaborated within a Scrum team to design, implement, test, and deliver the system within **4 weeks**.  
**Coding Instructor**, Pioneer Academy – Egypt (Remote) Feb 2026 – May 2026  
• Explained courses to college students, such as Regression, OOP, and problem-solving.  
**Research Intern**, Nile University - Cairo, Egypt (Hybrid) Jun 2025 – Sep 2025  
• Conducted data preprocessing, EDA, and geospatial data analysis on the NetMob 2025 dataset (~**500 million rows**) using PySpark and Kepler.gl, Uncovered the most optimal locations for the installation of piezoelectric materials in Île-de-France.  
• The research paper was presented as a poster at the **NetMob 2025 conference in Paris**.  
**Big Data trainee**, Samsung Innovation Campus - Cairo, Egypt (Hybrid) Jul 2024 – Sep 2024  
• Learned SQL, Hadoop ecosystem, Cassandra, MongoDB, Spark, and more about Big Data and Data Visualization.

## Projects

**PavementEye | Road-Cracks detection System** [GitHub Repo](#)  
• Fine-tuned a computer vision model (YOLOv8s) on the **EGY\_PDD research dataset**. Developed a robust **streaming data pipeline** using Apache Kafka, Spark, Cassandra, and Docker, and deployed the system on Azure. Integrated detections with **OpenStreetMap** to enrich **geospatial analysis**.  
• Processed **1 image/second** and detected **100+** pavement cracks across **11 crack types** in **under 30 minutes**.  
• Top **50 finalists** in the Huawei Developer Competition 2025 – Northern Africa.  
**Automated OT/ICS Threat Intelligence Agent** [GitHub Repo](#)  
• Built an automated pipeline to extract OT/ICS CVEs from the NVD API every 10 minutes using **two-stage filtering**: heuristic keyword matching followed by reasoning with a local **LLM (Llama 3.2)**, generating actionable risk insights.  
• Classified **69 OT/ICS CVEs** and developed a Streamlit dashboard with impact/exploitability visualizations and advanced filtering.  
**Exam Timetabling for Buraydah Colleges | Graph Coloring** [GitHub Repo](#)  
• Worked on real data from Buraydah Colleges to solve the exam scheduling problem using **graph coloring**.  
• Successfully distributed **1820 Students** across the specified room capacities (**18 rooms** with **2190 seats**), **6-time slots**, and **496 registered courses** with no conflicts in the minimum time needed (**8 days**) after optimizing time with resources.

## Skills

### Technical Skills:

**Languages:** Python, Java, C++ , SQL, Bash, JavaScript.

**AI & Data:** Hadoop, NumPy, Pandas, GeoPandas, Matplotlib, Seaborn, Plotly, Scikit-learn, PySpark, SSIS, Apache Kafka, Power BI, Apache Airflow, Apache NiFi, Streamlit.

**Other Tools:** Kepler.gl, Docker, Git & GitHub, LaTeX.

**Software fundamentals:** OOP, Data Structures & Algorithms, Databases(SQL & NoSQL), Computer Networks, Cloud Computing, Operating Systems, System Design.

**Soft Skills:** Problem Solving, Team Leadership, Planning, Presentation Skills, Research Skills.

## Achievements and Certifications

• SprintUp – AI and Machine Learning Foundations, Sprints 2025  
• Ranked **5th** within faculty, **25th** overall on contest day, ECPC 2024  
• Python for Data Analysis, NTI 2024

## Volunteer work

**Manager of Projects**, IEEE ALEX CSC Jul 2026 – Present  
• Built an automated open-source coding challenge platform using **Go**, **PostgreSQL**, and **GitHub Actions CI/CD pipelines**.  
**Problem Solving Trainer**, ICPC FCDS Dec 2024 – Jun 2025  
• Facilitated **275 Students** in competitive programming by creating **5 video explanations** for problems and answering their questions.  
• Created **6 problems** on polygons using LaTeX for contests and sheets on Codeforces.  
• Explained sets, maps, and priority queues in C++ to students in an offline session.